

Article

Development of Parametric Prostheses for Different Levels of Human Hand Amputations Manufactured Through Additive Manufacturing

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Abstract: Upper limb prostheses face acceptance challenges due to factors such as discomfort, limited functionality, high weight, and elevated costs. Despite the availability of advanced models with sophisticated technologies, their accessibility remains limited to individuals with greater financial means. This study presents the development of a parametric hand prosthesis designed for total or partial amputations, utilizing additive manufacturing and affordable hypoallergenic materials. The parametrization was based on customized anthropometric geometries, modeled using CAD software, with structural validation conducted through finite element analysis under static forces. The printing material was characterized according to ASTM D638 standards. The results indicate that the prosthesis provides a personalized, functional, and accessible solution that meets the specific needs of users. It is concluded that the use of additive manufacturing and parametric design not only reduces costs and improves accessibility but also enables the creation of devices tailored to individual user characteristics, promoting greater functionality and quality of life.

Keywords: prosthesis; additive manufacturing; parametrization; amputation; finite element analysis



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1. Introduction

Considered one of the main elements of the human locomotor system, the hands enable movement and object manipulation. They are composed of muscles, tendons, nerves, blood vessels, and bones. The hands are seen as one of the most complex and versatile parts of the human body. Located at the end of the upper limbs, they are essential for motor behaviors while also allowing the expression of social and emotional behaviors through gestures and touch, as well as artistic expressions, such as in the arts and music [1,2]. Recent studies indicate that the functional complexity of the hands is directly related to their ability to perform a wide range of movements and interactions with the environment [3,4].

Thus, they have the ability to perform various critical functions essential for daily life. Therefore, the partial or total loss of the hand leads affected individuals to seek, as an

alternative, a permanent substitute for the missing region through the use of orthopedic devices such as prostheses, with the aim of restoring the functionalities of the partially or completely lost hand. These prostheses are designed to mimic the appearance and function of the human hand, with several types available, including passive, biological, mechanical, and bionic prostheses [3–5].

Advancements in additive manufacturing technologies have greatly enhanced the customization of prostheses, enabling better adaptation to individual users and improving functionality to more closely resemble natural movement [6,7]. This technology also allows for the creation of prostheses with intricate geometries and a high degree of personalization, offering a substantial improvement over conventional manufacturing methods [6–9].

The functionality of hand prostheses is closely linked to the level of technology incorporated into their design. Simpler prostheses, such as mechanical ones operated by muscle propulsion, are limited in their range of motion, typically restricted to basic actions such as opening or closing the hand in a pincer grip. Recent studies highlight that while mechanical prostheses are still useful, they face significant limitations in terms of movement control and precision, while bionic prostheses offer significantly improved performance [4,10]. Bionic prostheses, which integrate advanced components such as microprocessors, boards, actuators, sensors, and control systems, enable users to replicate hand movements that closely resemble natural motions, allowing for fast and precise movements to grasp and hold objects. This advancement has increased user confidence and control in daily interactions [10,11]. Additionally, advancements in neuroprosthetics, which integrate complex sensors into the user's muscles to capture bioelectric signals, have enhanced the precision and control of upper limb prostheses [12].

The loss of a limb or part of it can be caused by a variety of reasons, including congenital causes, injuries, diseases, infections, and neoplasms. Among these causes, the method used for the total or partial removal of the limb is amputation. Amputation, considered one of the oldest surgeries in medicine, continues to be one of the leading causes of disability in the upper limbs. Amputation of the upper limb, especially the hand, usually occurs during the active age due to traumatic or physical causes, work-related thermal injuries, or injuries sustained in military combat [13,14].

Individuals with disabilities face significant challenges in performing daily activities. Thus, one of the main goals of rehabilitation is to reintegrate the individual mentally and physically into their new reality, that is, their new mental and physical state. In this context, adaptation and training for the use of orthopedic devices, such as prostheses, have proven to be essential for optimizing the functionality and potential of these devices, which can significantly improve patients' quality of life [15–17]. Studies like this show that functional training for prosthesis use can speed up the adaptation process and improve user performance in daily activities.

However, significant problems are currently presented by conventional hand prostheses, arising from issues such as low customization, resulting in an aesthetically unappealing appearance [18,19]. Issues such as dimensional inaccuracies, excessive weight, and discomfort during use are critical factors that directly affect the functionality of prostheses. Additionally, limitations in manipulation and overall convenience further undermine their effectiveness [18,20].

An upper limb prosthesis considered “ideal and comfortable” should be recognized by the user as an extension of their natural body. It must replace and meet, within its potential and functionalities, the motor, emotional, and social capabilities of the amputated limb, promoting practical and pleasant use [15,21]. Creating devices that take into account the anatomy and individual preferences of the patient can increase the acceptance and efficiency of prostheses [17].

This article aims to develop a parametric hand prosthesis designed to address various levels of partial or total hand amputations using additive manufacturing technologies. The proposed approach leverages advanced manufacturing methods that allow for the customization of the prosthesis according to the anatomical features and functional needs of each patient. The primary focus of this work is on the characterization of materials, design development, construction, and additive manufacturing of the prosthesis, with a particular emphasis on engineering aspects such as dimensional accuracy, optimization of manufacturing processes, and sustainability. This project aligns with recent advances in the application of additive manufacturing for prosthesis customization [6–9].

Furthermore, the mechanical feasibility of the prosthesis will be assessed through finite element analysis (FEA), which will help ensure the prosthesis's structural integrity and functionality at various amputation levels. FEA serves as a critical tool for predicting the mechanical behavior of the prosthesis prior to physical manufacturing, enabling design optimization and the identification of potential weak points. While clinical testing with patients is vital for the final validation of the prosthesis, this stage is not part of the present study. However, it is planned as a future step in our multidisciplinary bioengineering laboratory, where the integration of various engineering and health disciplines will be key to the next phase of development.

2. Methodology

For this work, five methodological steps were defined: (1) characterization of materials; (2) development and design of the virtual model of the device; (3) parametrization of the mechanical prosthesis to allow personalized adjustments; (4) execution of computational simulations using Finite Element (FE) analysis on the device; and (5) printing and construction of the mechanical hand prosthesis through Additive Manufacturing (AM).

2.1. Material

This study focuses on Tritan, a copolyester polymer widely used in filament form for 3D printing via FDM (Fused Deposition Modeling) [19]. With a diameter of (1.75 ± 0.05) mm, the filament ensures consistency and is compatible with most standard 3D printing systems. Tritan is well-known for its impressive mechanical and thermal properties, such as high tensile strength, impact resistance, and rigidity, which make it an ideal choice for producing durable, functional parts like prosthetic components.

One of the notable characteristics of Tritan is its ability to remain thermally stable, with a high glass transition temperature (T_g) of approximately 100–110 °C, allowing the material to withstand high temperatures without losing its structural integrity. This is crucial for applications where the material may be exposed to varying thermal conditions. Additionally, Tritan's melting temperature typically ranges between 230 and 250 °C, providing reliable fusion during the FDM printing process.

Unlike other materials, Tritan is not brittle, which significantly improves its durability, making it suitable for designs that require flexibility and strength. Its high resistance to chemical agents, such as oils, solvents, and common cleaning solutions, further contributes to its suitability in environments where exposure to such substances is frequent. This characteristic is particularly important in the context of prosthetic applications, where hygiene and long-term wear resistance are key factors.

Tritan also exhibits transparency, which can be an advantage when designing prosthetic devices that seek a balance between functionality and aesthetics. In orthopedic applications, Tritan's ability to combine strength, flexibility, and chemical resistance makes it viable for the design of parts with complex geometries, often required in customized prosthetic designs, offering both form and function. Furthermore, the material's ability

to be printed with high precision in standard FDM 3D printers supports its viability for manufacturing custom prosthetic parts with dimensional accuracy and efficiency.

Due to these combined properties, Tritan has shown great orthopedic potential, particularly for functional designs, such as hand prostheses, where the balance between durability, comfort, and adaptability is essential for patient satisfaction and device longevity.

2.2. Equipment

The Brazilian-made GTMax 3D Core A3 FDM printer (Lab Maker Living Lab MS, Campo Grande, Brazil) was chosen for meeting industrial and professional demands, offering a print volume of $300 \times 300 \times 300$ mm, ideal for producing large parts or multiple pieces in a single run. Its CoreXY structure provides high precision and speed, with a layer resolution ranging from 50 to 300 microns. Compatible with engineering filaments such as PLA (Polylactic Acid), ABS (Acrylonitrile Butadiene Styrene), PETG (Polyethylene Terephthalate Glycol), and Tritan, among others, and equipped with a hot-end capable of reaching up to 300°C , this printer allows the use of advanced materials. The adjustable heated bed, reaching up to 110°C , improves adhesion and minimizes warping. Additionally, features such as print recovery and a filament sensor enhance process reliability [22].

The tensile mechanical tests were conducted at the Materials Strength Laboratory of the Pontifical Catholic University of Minas Gerais (PUC-MG). For the experimental procedures, a universal testing machine, model 23-200 Series 23 EMIC (Instron Brasil Equipamentos Científicos Ltda, São José dos Pinhais, Brazil), was used, featuring a maximum displacement speed of up to 500 mm/min and a load cell with a capacity of 200 kN. The equipment is factory-calibrated in accordance with the ABNT NM ISO 7500-1 standard and is certified under Certificate No. 18080801MC. The machine is controlled by the Tesc software, version 3.04. During the test, the displacement rate used was specified by ASTM D638-14, equal to 5 mm/min. Initial tests were performed on five test specimens (Cps) at the same rate of 5 mm/min to conduct a preliminary analysis of the equipment. The strain measurement of the test specimen (Cp) was carried out using the extensometer integrated into the equipment.

2.3. Normalization

For the purpose of constructing and evaluating the mechanical properties of the material used in three-dimensional printing, the ASTM D638—Standard Test Method for Tensile Properties of Plastics—was applied for the fabrication of the test specimens [23]. The dimensions considered for the design of the standardized test samples for the tensile test are specified in the aforementioned standard, along with the methodological recommendations for conducting the tensile test. For the preparation of the test specimens, the preferred sample dimensions, Type I, were selected—Figure 1.

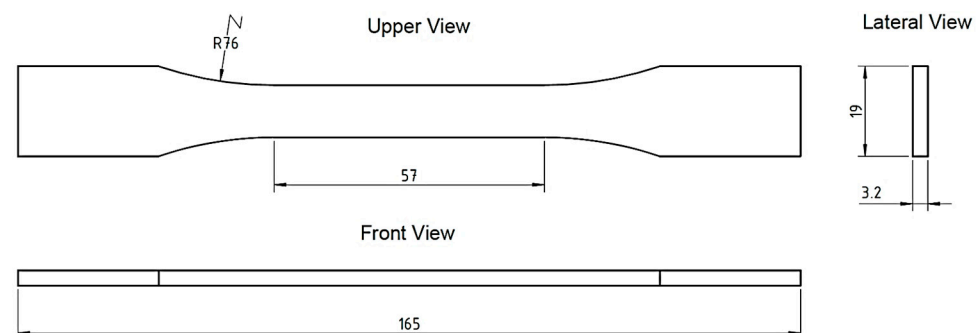


Figure 1. Specimen dimensions (in millimeters).

The printing of the test specimens was carried out in compliance with the ASTM 52921—Standard Terminology for Additive Manufacturing—Coordinate Systems and Test Methodologies [24]. This standard includes terms and nomenclature related to systems, coordinates, and test methodologies for Additive Manufacturing technologies. It establishes requirements for validating the printing process, illustrating the location and orientation of the parts through the build planes that can be followed during the three-dimensional printing process, as shown in Figure 2.

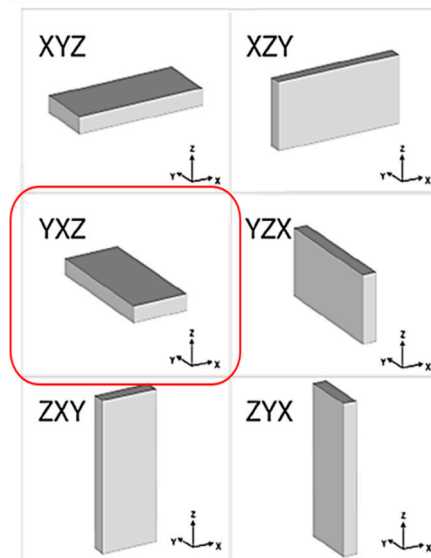


Figure 2. Printing plane YXZ.

The constructive plane used for the fabrication of the test specimens was the YXZ plane, as shown in Figure 2. This choice was made because the XYZ and YXZ printing direction references exhibit better mechanical strength while achieving good dimensional accuracy in the printed part using FDM technology [25–27].

In the manufacturing of parts through 3D printing, it is essential to determine the key operating parameters after selecting the material to be used in the process. This includes defining the type of internal infill and its density, factors that directly influence the reduction of printing time and material consumption. All printing parameters employed are detailed in Table 1.

Table 1. Printing Parameters.

Parameters	Value	Units
Printing Plan	YXZ	-
Infill	Tri-hexagonal	-
Speed	40	mm/s
Nozzle Temperature	265	°C
Bed Temperature	110	°C
Shell	3	perimeters
Layer Height	0.2	mm
Top Layer	4	mm
First Layer	0.2	mm

To ensure that the tensile test achieves real and reliable results, it is of utmost importance that the printing of the test specimens and the execution of the mechanical test follow standardized methodologies.

Thus, for the preparation of the test specimens and the execution of the destructive test, the infill density was considered a variable parameter, using the Tri-Hexagonal/Honeycomb pattern (Figure 3), generated through the Ultimaker Cura 5.8.1 or Orca Slicer software 2.1.1.

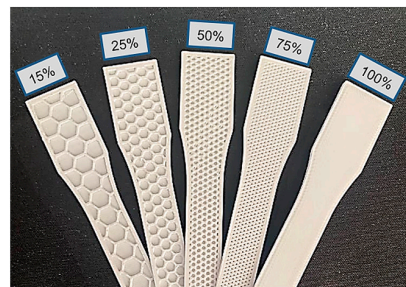


Figure 3. Distribution of the internal geometry of the test specimens at 15, 25, 50, 75, and 100%.

This type of infill is considered one of the patterns that provide greater mechanical strength to parts printed using FDM technology. Its hexagonal geometry follows a geometric pattern that not only offers increased resistance in all directions and good printing speed but also enhances the balance between strength, cost, and printing time. Additionally, it stabilizes and homogenizes the layer deposition process, preventing printing defects such as warping, which is the deformation of the part caused by material contraction that can completely compromise the print [28,29].

For the printing of the test specimens, the following parameters were considered as fixed: (1) shell: the layer responsible for the external “wall/shell”, which directly influences the lateral protection and the aesthetic appearance of the printed part; (2) top layer: refers to the upper closing layers, with the objective of finishing the top surface of the part; (3) bottom layer: refers to the lower closing layers, characterized as the first layers printed, providing support and protection to the base of the part; (4) speed: a parameter related to printing time and adhesion between layers; (5) temperature: responsible for the extrusion of the filament during 3D printing [30].

Figure 4 illustrates some of the fixed parameters used in this work.

2.4. Experimental Test

For a better understanding of the methodology for conducting the tensile test, Figure 5 presents a brief flowchart to generically illustrate the methodological steps of analysis through the tensile test and scanning electron microscopy.

In this experiment, the material properties evaluated are (1) Maximum Stress (Tensile Strength), which is the highest stress the material can withstand when subjected to tension, without showing signs of internal or external fracture in the test specimen; (2) Yield Strength or Yield Stress (0.2%), which marks the onset of the yielding phenomenon and the transition to plastic deformation, that is, the point at which permanent deformation begins in the test specimen; and (3) Modulus of Elasticity or Young’s Modulus, a mechanical property that indicates the stiffness of a solid material, meaning that the higher the modulus of elasticity, the smaller the resulting elastic deformation when a given stress is applied [31].

Using the initial form of the TRITAN filament and after defining the printing parameters, test specimens (cps) were generated and divided into five different groups (G1, G2, G3, G4, and G5) with five distinct levels of internal infill, specifically: 15%, 25%, 50%, 75%, and 100%. A total of 25 test specimens were produced, subdivided into 5 groups, as follows: (1) quintuplicate with 15% infill; (2) quintuplicate with 25% infill; (3) quintuplicate with 50% infill; (4) quintuplicate with 75% infill; and (5) quintuplicate with 100% infill.

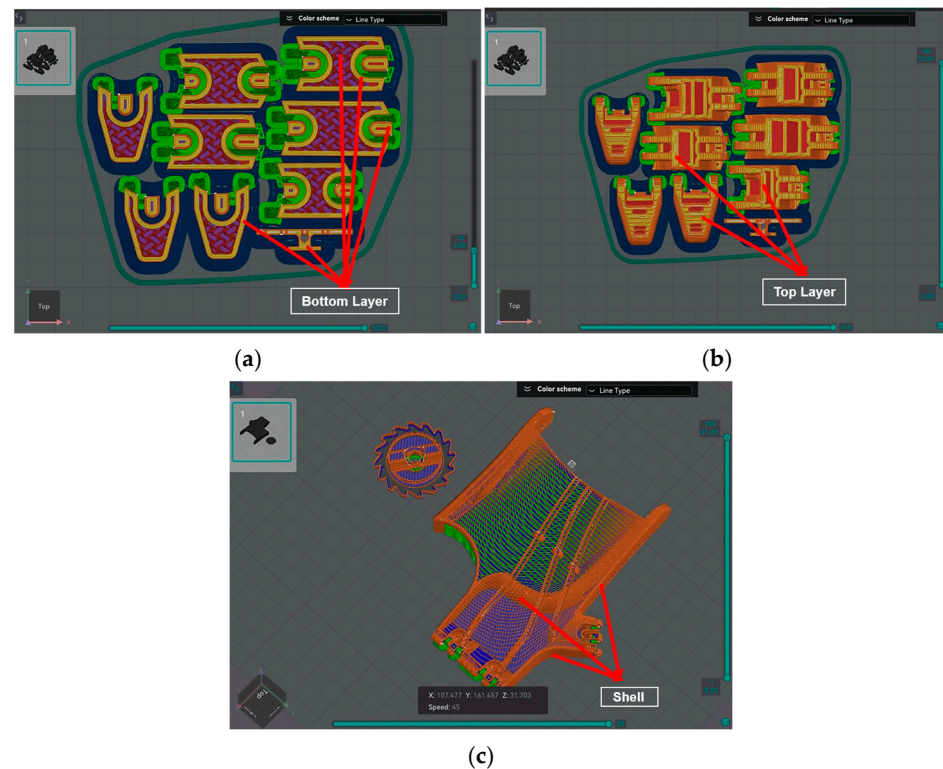


Figure 4. Fixed parameters: (a) bottom layer, (b) top layer, and (c) shell.

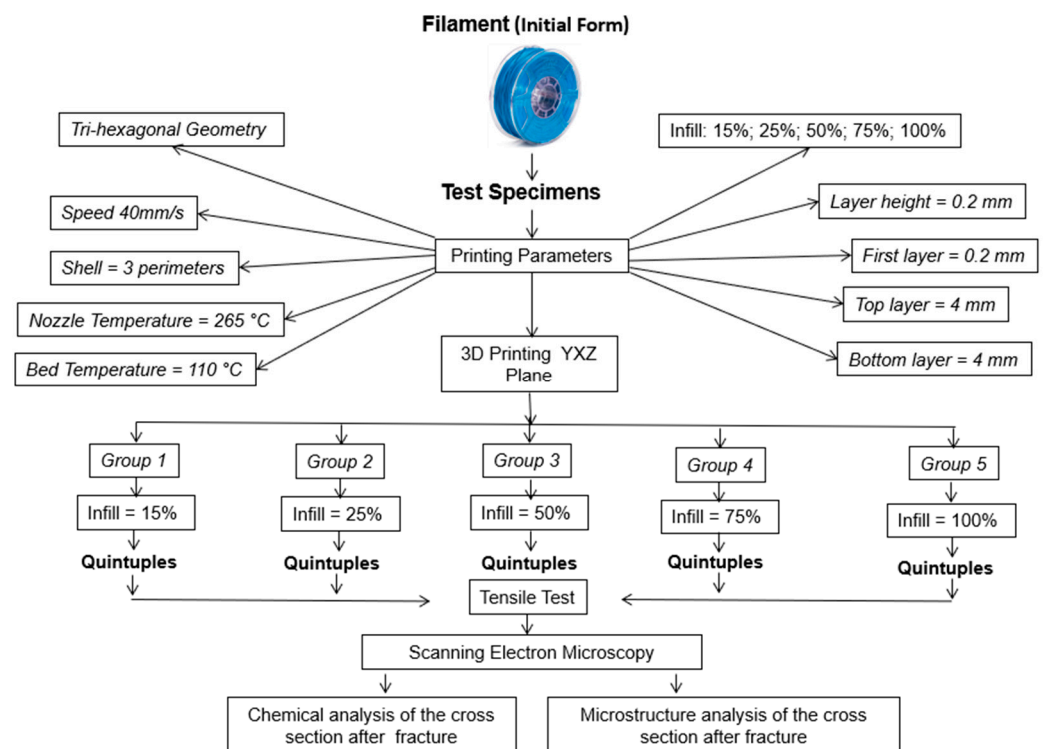


Figure 5. Flowchart of the experimental methodology.

2.5. Development of the Hand Prosthesis Parameterization

Figure 6 schematically illustrates the steps involved in the research methodology of this work, focusing on the parametrization of the hand prosthesis.

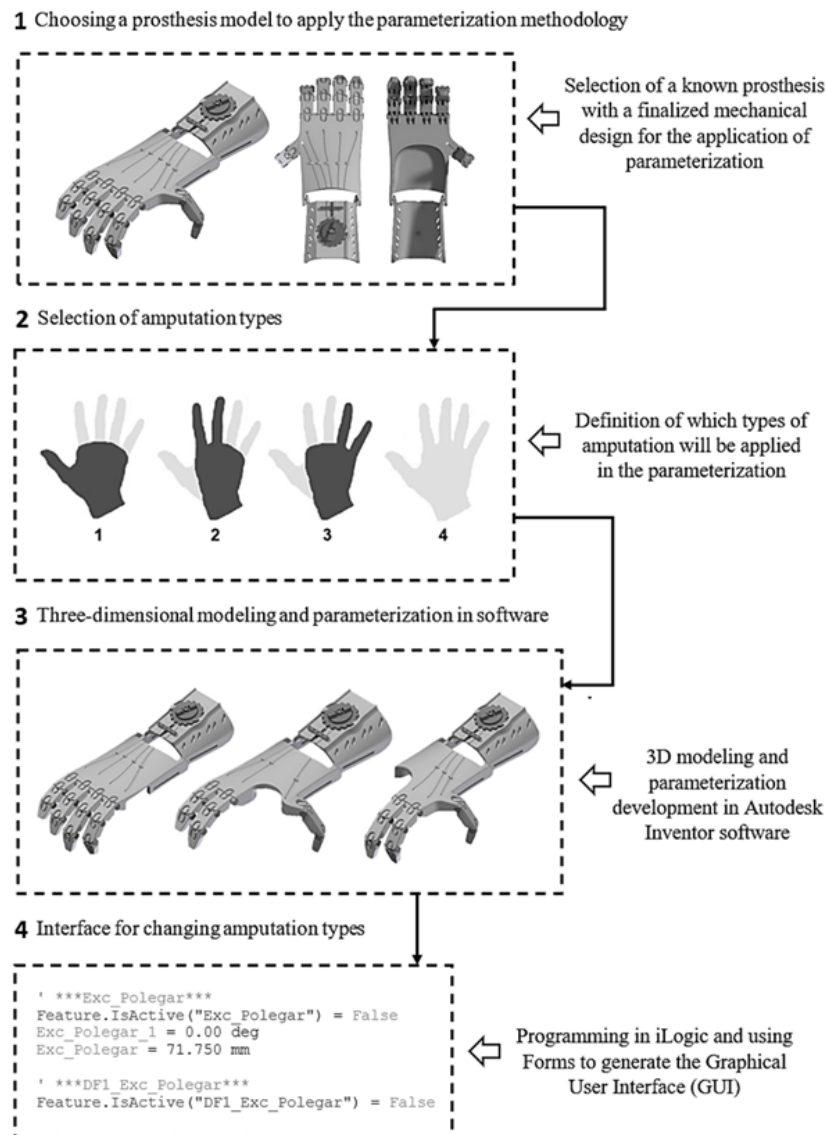


Figure 6. Schematic diagram of the applied methodology.

The parametric methodology consists of applying parametrization techniques with the objective of adapting the prosthesis for four different types of hand amputations, as shown in the following section.

The parametrization process is justified by the 3D CAD design of the Esperança prosthesis, which features a robust and functional design, allowing customized modifications according to the users' anatomical characteristics. The research will follow a structured approach in three main steps: (1) detailed analysis and refinement of the initial prosthesis model, identifying critical parameters that influence its performance; (2) definition of the three types of human hand amputations; and (3) development of a parametrization process to be applied to the prosthetic model, with the ability to adjust variable parameters, aiming to modify dimensional aspects for different profiles of total or partial hand amputations using the 3D CAD base as a development and adjustment platform.

This methodological approach aims to contribute to the improvement of customized prosthetic solutions, with the potential to expand the scope of future applications of the Esperança prosthesis, enabling its viability in the field of rehabilitation for amputee patients.

2.6. Selection of the Prosthetic Device

The present research is based on the hand prosthesis named *Esperança Labbio*, illustrated in Figure 7, developed at the Bioengineering Laboratory (Labbio) of the Federal University of Minas Gerais (UFMG) and enhanced using 3D CAD modeling.



Figure 7. Prosthesis *Esperança Labbio*.

The device was chosen as the basis for applying a parametrization methodology aimed at different types of hand amputations, as shown in Figure 6. This choice is justified by its comprehensive and functional mechanical design, which allows adaptations according to the individual characteristics of each patient.

2.7. Selection of Hand Amputation Types

The selection of Transmetacarpal, Partial Radial, Partial Ulnar, and Total amputations for prosthesis parametrization is justified by their representation of different degrees of functional loss and biomechanical challenges. Each type provides a basis for customized solutions, ranging from partial hand preservation to complete removal, enabling the prosthesis to be adapted to the specific needs of the remaining functionality.

The Transmetacarpal amputation preserves the thumb and part of the palm, requiring a focus on grasping functionality; the Partial Radial amputation, involving the loss of the thumb and two fingers, prioritizes the recovery of pinching ability and firmness; the Partial Ulnar amputation compromises fingers essential for precision, demanding adaptations for fine tasks; and the Total amputation removes the entire hand, necessitating solutions that promote greater independence. These levels of amputation provide a comprehensive approach for developing parametrized prostheses applicable to various clinical conditions.

2.8. Parameterization Method by Suppression

For the four types of amputations, parametrizations were developed using the geometric suppression method, which consists of the controlled removal of parts from an existing three-dimensional solid or changing the geometric and dimensional parameters, adjusting them to the specific needs of each design. This approach was applied both to the modeling of the palm and mechanical assemblies, allowing for the exclusion of unwanted components and the creation of customized configurations.

In the case of the palm, carefully designed sketches were used to perform the suppression of the desired geometry, complemented by subsequent commands that finalized the three-dimensional model. Figure 8 presents the three-dimensional model of the palm, the main parametrized component in this process, standing out as the foundation for customizations that ensure the ergonomics and functionality of the prosthesis.

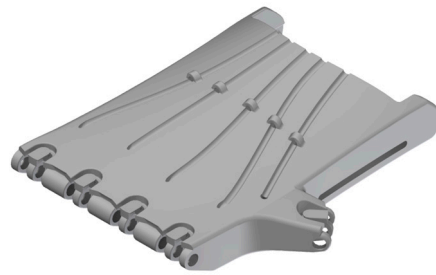


Figure 8. Developed parametrized palm.

In the context of the three-dimensional assembly of the prosthesis, parametrization through suppression plays a crucial role by removing or adding fingers according to the specific type of amputation selected by the user, as well as modifying the assembly screws of the set.

2.9. Parameterization and Analysis of Level 3 Hand Amputation

The application of parametrization, the study of mechanical feasibility through finite elements, and the construction of the prosthetic device were carried out for all four levels of hand amputation. However, in this paper, the results for level 3 are presented, which involves the amputation of the thumb, index, and middle fingers. The choice of this level is due to its functional complexity, representing one of the most challenging cases in terms of loss of capacity and prosthesis adaptation. Although the other levels were also analyzed, the space limitation allowed for a focus only on this specific level, enabling a more detailed and representative approach to the proposed solutions for partial hand amputations.

Figure 9 presents the initial sketch for suppression in the palm model. The extrusion command, named *Exc_Ane_Min_1*, shown in the figure by the command tree, refers to this sketch, considering a user who has only the ring and pinky fingers.

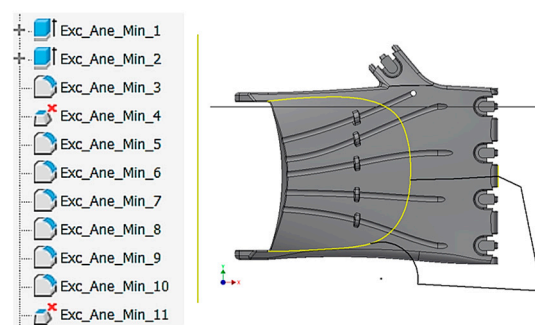


Figure 9. *Exc_Ane_Min_1*: Suppression of the ring and pinky fingers.

In the lower region of the palm, the extrusion called *Exc_Ane_Min_2* was created based on the sketch in Figure 10, forming the palm housing and the opening for the user's ring and pinky fingers. *Exc_Ane_Min_3*, shown in Figure 11, creates the concavity radius in the palm, providing smoother geometry to the previous base design. This process is finalized by the fillet commands *Exc_Ane_Min_4* to *Exc_Ane_Min_10* (Figure 12). Finally, the delete face commands *Exc_Ane_Min_4* and *Exc_Ane_Min_11* are responsible for removing the support for the actuation wire of the ring and pinky fingers (Figures 13 and 14), which are not applied to the prosthesis model.

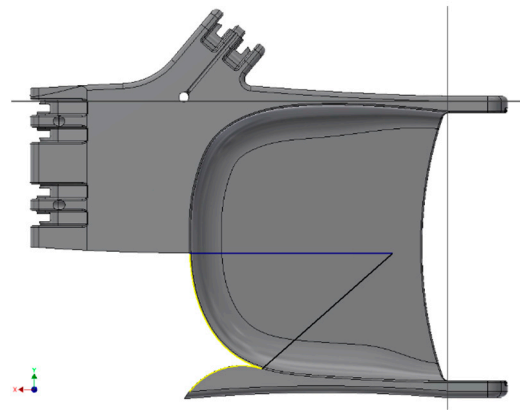


Figure 10. Exc_Ane_Min_2: Base sketch of the extrusion in the palm Type 3.

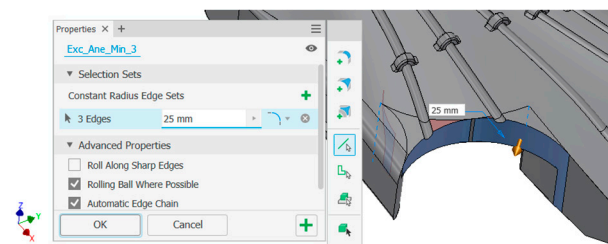


Figure 11. Exc_Ane_Min_3: Palm concurrence radius.

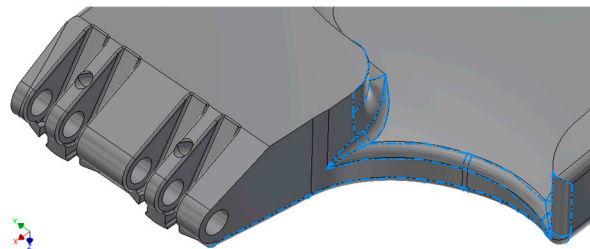


Figure 12. Exc_Ane_Min_4 and Exc_Ane_Min_10: Palm smoothing radius.

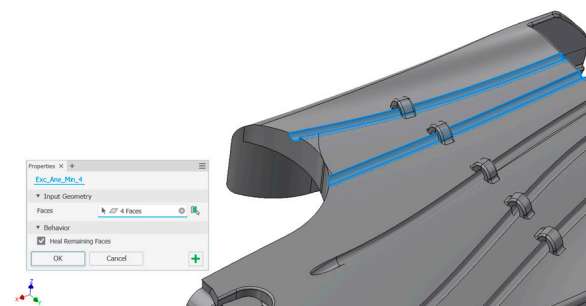


Figure 13. Exc_Ane_Min_4: Removal of the actuation wire path.

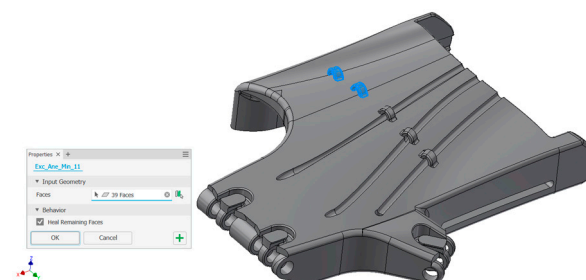


Figure 14. Exc_Ane_Min_11: Removal of the actuation wire support.

2.10. Mechanical Feasibility Study

In alignment with the objectives of this work, a finite element analysis was performed to represent the behavior under static load forces, considering the hand prosthesis developed and fabricated through additive manufacturing. Studies like [32] provide an important theoretical foundation for understanding elastoplastic contacts in rigid materials, contributing to the accurate modeling of mechanical structures. This type of approach allows for the evaluation of stress and strain distribution, optimizing the device's performance in real-world usage scenarios.

This analysis was performed through computational simulation of static stresses, aiming to evaluate the mechanical responses of the prosthesis under the application of compression forces when gripping an object. The execution of the simulations involved important steps such as geometric modeling, boundary conditions and acting forces, boundary constraints, reduction and adaptation of the prosthesis model to create the mesh of the object, defining the materials that compose the prosthesis, and determining both the conditions for the fixation of the applied forces (elements and/or regions of application) and the acting forces themselves on the object.

The three-dimensional model of the hand prosthesis was initially obtained in the ".stl" format, a widely used standard for 3D printing, which represents the model's surface through a triangular mesh. However, for the simulations to be conducted, it was necessary to convert this mesh into a structure more suitable for computational analysis.

Initially, the triangular mesh was converted into a quadrilateral mesh (quad mesh), which offers advantages in terms of element regularity, reducing distortions and facilitating the application of local refinements where higher stress gradients are expected. The conversion to quad mesh was performed in a way that preserved the geometric details of the prosthesis, ensuring that the critical parts of the structure, such as the finger joints and the region of connection to the arm, maintained their structural integrity.

Next, the quadrilateral model was transformed into a parametrized solid body in the ".f3d" format, allowing for more direct manipulation of material properties and precise definition of boundary conditions. This solidification process is essential for the simulation to account for the internal distribution of stresses, enabling a more realistic assessment of the prosthesis's mechanical responses. Additionally, this parametrization facilitates the adaptation of the model to different configurations and loads, making the study more flexible.

Figure 15 illustrates the mesh transformation process: (a) shows the model in triangular mesh format (.stl), while (b) presents the model after conversion to a solid body in ".f3d" format, highlighting the increased regularity of the mesh and the accuracy in the geometric representation of the prosthesis. The highlight in Figure (b) illustrates the refinement applied to the region with the highest concentration of stresses, allowing for a detailed analysis of this critical area.

2.10.1. Boundary Conditions and Acting Forces

The prosthesis operates based on the tenodesis effect, which serves as its primary functional principle. The pre-tension in the prosthesis traction cables, activated by wrist flexion and extension, generates a cylindrical pinch movement induced by finger flexion. Passive extension is achieved through orthodontic elastics or nylon threads located on the dorsal region of the fingers, with fixation on the palmar region of the prosthesis.

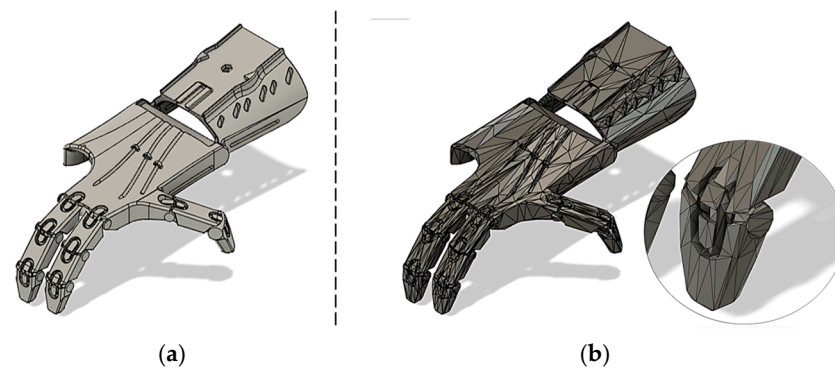


Figure 15. Mesh transformation: (a) mesh model .stl extension, (b) parametrized model in body format .f3d extension.

The fixation mechanism is established by stabilizers modeled on the residual limb and adjusted with directional tensioning ratchets, ensuring stability and proper tensioning of the prosthesis cables, which function as flexor and extensor tendons. This design allows the prosthesis to replicate natural grasping and releasing motions effectively.

The main acting forces and constraints were established based on a review of the scientific literature on upper limb prostheses, also considering the typical functional use characteristics (Figure 16).

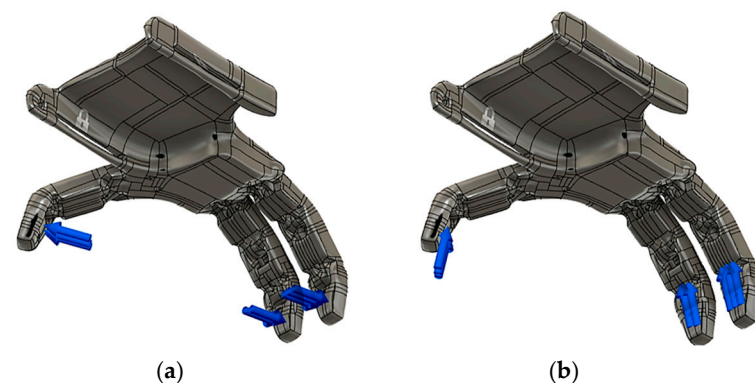


Figure 16. Force allocation: (a) gripping force, (b) friction force.

Gripping Force: The force applied at the fingertips to simulate the action of gripping an object was defined as 50 N, distributed uniformly along the contact surfaces of the fingers. This value aligns with studies indicating that the average gripping force in daily activities ranges between 40 N and 60 N for adults using upper limb prostheses [33]. The choice of this value represents an intermediate condition, suitable for light tasks, such as holding a cup or other household items.

Friction Forces: A static friction coefficient of 0.4 was adopted for the contact surfaces of the fingers, aiming to simulate interaction with objects made of materials such as glass and lightweight plastic. This value was based on studies on polymer surfaces [34] and ensures that the prosthesis is capable of securely and efficiently holding and manipulating objects.

2.10.2. Boundary Constraints

The boundary conditions were defined to simulate the interaction of the prosthesis with the user's residual limb, ensuring the necessary stability during use. These conditions consider the forces and natural movements expected in daily activities, ensuring that the analysis accurately reflects the real-world usage conditions of the prosthesis.

Base Constraints: The base of the prosthesis, which establishes the connection between the prosthetic component and the user's forearm, was fully fixed. This fixation was designed to represent the anchoring interface between the prosthesis and the residual limb, mimicking the rigid and stable fixation expected in clinical practice. Figure 17 illustrates the areas and fixation points used, where all translational and rotational components were restricted to prevent unwanted movements that could compromise the functionality of the prosthesis.



Figure 17. Allocation of constraints.

The choice for a rigid fixation of the base is based on studies that highlight the importance of a stable interface to ensure that the prosthesis movements are transmitted efficiently to the user [35]. The stability of this interface is essential to prevent misalignments and energy losses, especially in activities that require precision, such as picking up small objects or holding containers.

The finite element analysis model was developed considering Tritan as an isotropic linear elastic material due to its mechanical properties that allow the application of this model with high precision. This type of material is widely modeled as an isotropic linear elastic material in studies involving finite element simulations, given that its mechanical properties exhibit uniform behavior regardless of the direction of load application. Although the 3D printing process by FDM commonly induces anisotropy due to the layer-by-layer deposition technique, it is reasonable to assume isotropic behavior for Tritan under the conditions considered in this study. This simplification is valid since the applied load (Gripping Force) is predominantly parallel to the printed layers, where the material exhibits relatively uniform mechanical properties [36–38]. The parameters used to define the material were entered into the modeling software, as illustrated in Figure 18, ensuring an appropriate representation of the material characteristics in the numerical model.

For the simulation, the model was configured using a quadrilateral mesh to ensure precision during the calculation of results. The regions with higher stress concentration were refined by increasing the density of mesh points, enhancing the accuracy of the analysis in critical areas. Regarding the load distribution, a uniform application was applied to the distal phalanges, simulating a compression scenario during the use of the prosthesis for holding lightweight objects.

Mechanical		
Young's Modulus	0.333 GPa	⬆️⬆️
Poisson's Ratio	0.03	⬆️⬆️
Density	1.320 g/cm ³	⬆️⬆️
Damping Coefficient	−0.02	⬆️⬆️
Strength		
Yield Strength	13.790 MPa	⬆️⬆️
Tensile Strength	100.000 MPa	⬆️⬆️

Figure 18. Mechanical properties of the analyzed material.

3. Results and Discussion

3.1. Specimen Measurements

Using a micrometer with a millesimal resolution (0.001 mm) from the brand DIGIMESS, the dimensions of the useful region of the printed test specimen (PB) were measured to verify if they met the width and thickness specifications established by ASTM D638. The measurements indicated that the width was 13 ± 0.1 mm and the thickness was 3.2 ± 0.3 mm. Five measurements were taken in each direction in the useful region, and the obtained values, along with their averages and deviations, confirmed compliance with the dimensions defined by the standard.

3.2. Tensile Test

Five tests were conducted for each group of samples with infill percentages of 15%, 25%, 50%, 75%, and 100%. Figure 19 illustrates the results of the samples that exhibited maximum stress values closest to the average of their respective group. Table 2 summarizes the results of the mechanical properties obtained from the tensile tests performed.

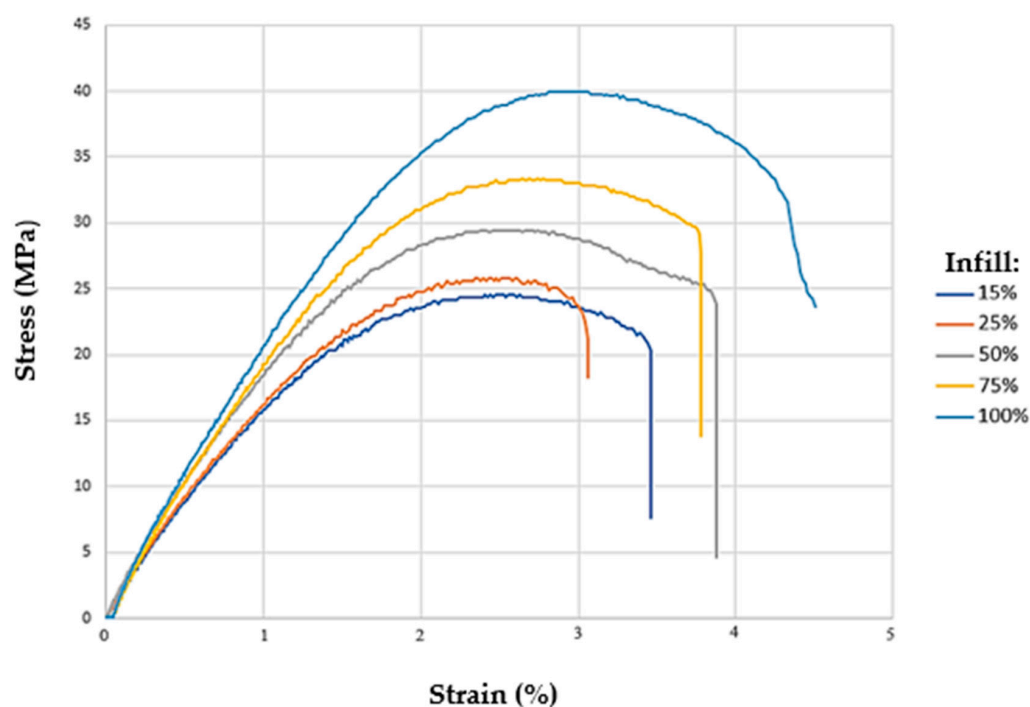


Figure 19. Engineering stress–strain curves for different infill levels of 15%, 25%, 50%, 75%, and 100%.

Table 2. Printing parameters.

Group	Strain Maximum (MPa)	Yield Stress (MPa)	Modulus of Elasticity (MPa)
1 (15%)	24.52 ± 0.28	12.90 ± 0.89	322.53 ± 17.17
2 (25%)	24.29 ± 0.22	13.79 ± 0.88	332.54 ± 7.76
3 (50%)	30.10 ± 0.55	16.33 ± 0.71	370.73 ± 9.78
4 (75%)	31.50 ± 0.22	18.86 ± 0.51	388.14 ± 18.72
5 (100%)	39.42 ± 0.19	20.08 ± 0.86	426.70 ± 16.62

From the results presented, a positive correlation can be observed between the infill level of the TRITAN samples and their mechanical properties, with increases in relative elongation, yield strength, tensile strength, and stiffness. It is important to emphasize that the stiffness values obtained from the slope of the stress–strain curve in the elastic regime, in this case, do not directly relate to the modulus of elasticity of TRITAN (estimated at approximately 1 GPa in printed samples) due to variations in the infill patterns used in the printing of each sample group.

The observed values for yield strength and tensile strength are also a reflection of the internal structure generated in each sample group. Higher infill patterns result in a greater effective cross-sectional area and, consequently, higher tensile strength. The highest infill level also gives the samples greater energy absorption capacity during deformation, as evidenced by the increase in tensile toughness and maximum elongation.

3.3. Scanning Electron Microscopy

The initial analysis of the fractured cross-section of the samples, after the tensile tests, revealed smooth displacements in the layers related to the printing conditions and material deposition. In each of the selected samples from the five tests conducted, with infill levels of 15%, 25%, 50%, 75%, and 100%, small displacements were observed along the lines corresponding to the internal infill and outer layers after rupture, as illustrated in Figures 20 and 21 for the 15% and 100% infill samples.

Additionally, a distinctive feature was observed in the final rupture region, where samples with higher infill percentages exhibited slight displacements relative to the central area of the cross-section. This variation in the rupture zone can be attributed not only to the application of a uniaxial load during the tensile test but also to the differences in internal infill density. The structural behavior of the samples was further analyzed using scanning electron microscopy (SEM), which provided detailed insights into the fracture surfaces.

The SEM images revealed notable differences in the mechanical performance between samples with 15% and 100% internal infill. In the samples with 15% infill, the fracture surface exhibited significant displacement between internal layers, which was accompanied by a localized concentration of stress at specific points. This led to cleavage and the initiation of cracks in certain regions, reflecting the brittle nature of the structure. These features are indicative of a less densely packed internal structure that is more susceptible to crack propagation under the applied uniaxial load. Consequently, the reduced infill percentage significantly compromises the material's overall tensile strength.

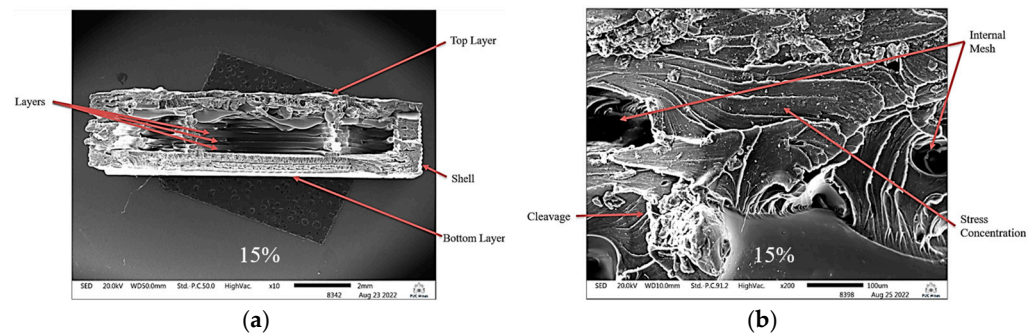


Figure 20. For the 15% infill sample, (a) shows the deposition lines of the layers, shell, and top/bottom layer; (b) shows cleavage, stress concentration, and internal lattice structures.

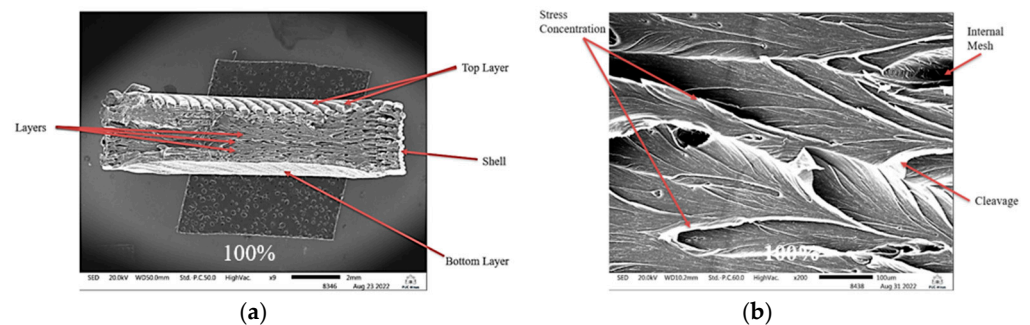


Figure 21. For the 100% infill sample, (a) shows the deposition lines of the layers, shell, and top/bottom layer; (b) shows cleavage, stress concentration, and the internal lattice structures.

Conversely, the samples with 100% infill demonstrated greater structural homogeneity, with minimal displacement between layers. The SEM analysis revealed that the stress distribution was much more uniform, contributing to a more robust material with a higher capacity to absorb deformation energy. The presence of a continuous internal structure also allowed for better stress transfer across the material, reducing the likelihood of localized failures and resulting in a more ductile and resilient behavior under loading conditions. These findings highlight the importance of infill density in optimizing the mechanical properties of 3D-printed materials and ensuring the structural integrity of the prosthetic components.

3.4. Three-Dimensional Model of the Generated Prosthesis

The modeling and parametrization of the prostheses were carefully developed using the constraints defined in iLogic, which played a key role in establishing a customizable structure for different types of amputations. This approach ensured that each prosthesis could be adapted to the unique needs of patients with various amputations. Specifically, the iLogic rules were applied to account for the different anatomical characteristics and limitations of each case. As a result, four distinct prosthesis models were generated to meet the specific requirements of the proposed amputation types.

Figure 22 illustrates the three-dimensional model of the Type 3 prosthesis, designed using the suppression rules for the ring and pinky fingers. These suppression rules allowed for the exclusion of unnecessary components, such as the fingers, ensuring that the prosthesis was optimized for the remaining parts of the hand. By implementing this process, the model not only became more functional but also more efficient in terms of material usage and user comfort.

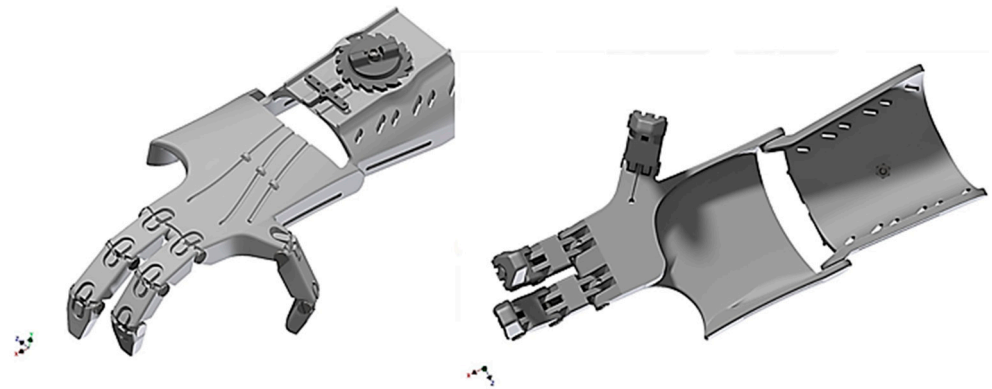


Figure 22. Generated Type 3 prosthesis.

3.5. Stress Distribution

The maximum von Mises stress obtained in the simulation presented localized values as a result of mesh interpolation with the formation of sharp edges, which generated a stress peak at the specific point analyzed. This phenomenon is characteristic of mesh configurations where abrupt transitions between elements cause elevated stress values in a localized manner.

The modeling of the system was configured considering adequate mesh refinement and appropriate application of boundary conditions to enhance the accuracy of the stress distribution analysis. The results obtained indicated the following maximum stress values at the three main critical points:

- Point 1—Lateral fixation region of the prosthesis: 12.276 MPa;
- Point 2—Metatarsophalangeal joint region of finger 2: 10.755 MPa;
- Point 3—Interphalangeal joint region (proximal/medial phalanx) of finger 3: 11.016 MPa.

The results demonstrate that the stress concentrations remain within acceptable limits, indicating adequate accuracy in the analysis performed.

The areas highlighted in blue represent regions with minimal or nonexistent forces, indicating areas of lower structural risk. Meanwhile, the areas in green and yellow tones show the points of highest stress concentration, especially at the joints and the fingertips, where direct contact with external surfaces occurs.

Figure 23 presents the color map corresponding to the stresses observed in the simulation.

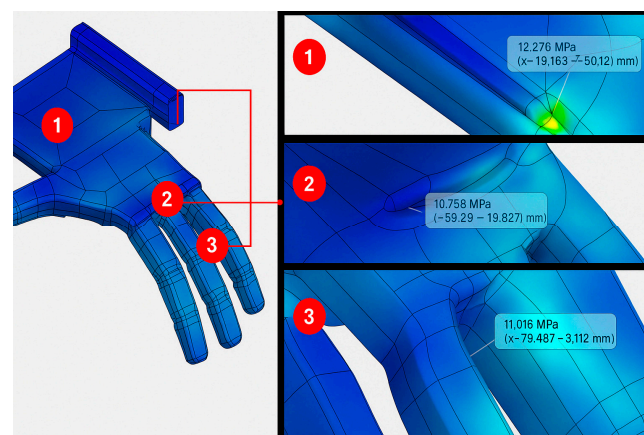


Figure 23. Color map corresponding to the stresses observed in the simulation.

The analysis suggests that the prosthesis structure is adequate to withstand typical gripping loads, but it may be necessary to reinforce the joints to enhance the device's

durability. Figure 24 presents the color map corresponding to the stresses observed in the simulation.

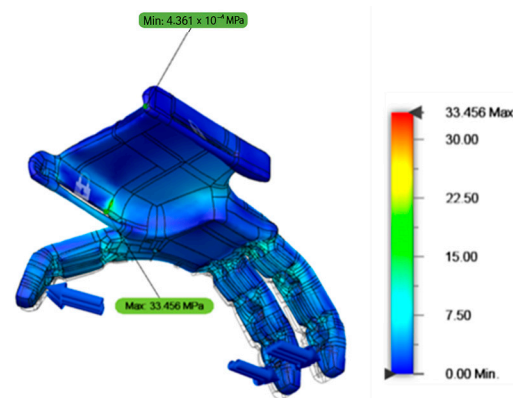


Figure 24. Color map corresponding to the stresses observed in the simulation.

3.6. Displacement Distribution

The displacement analysis of the hand prosthesis model reveals a maximum deformation of 1.68 mm, as observed at the fingertips (Figure 25). This displacement indicates that the prosthesis undergoes deformation under the applied load conditions, particularly in the areas farthest from the fixation, where the structure is subject to greater bending.

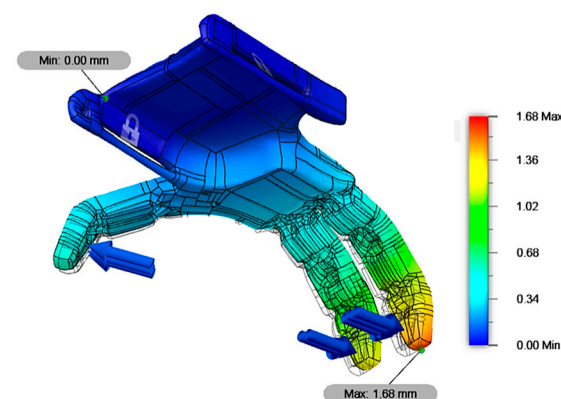


Figure 25. General color map maximum deformation.

The red areas in the image highlight the regions of greatest displacement, indicating the most critical areas, while the blue areas represent minimal displacements, with values close to 0.00 mm, showing stability in the fixed parts that are less subjected to load.

3.7. Device Printing

The developed prosthetic device features an anthropomorphic design and utilizes the tenodesis effect to generate movement through wrist flexion, complemented by passive finger relaxation driven by elastic components. The fingers and thumb are attached to a palm component with mounting screws, while the palm is articulated to a forearm stabilizer support in the shape of a “glove”.

The active flexion of the wrist is achieved through a high-strength non-elastic line, while the passive extension of the fingers is ensured by elastic components, such as orthodontic elastics positioned on the back of the fingers. The device was fabricated using FDM 3D printing technology, employing Tritan polymer material. With the 3D CAD (Orca Slicer software 2.1.1) design previously defined and the printing parameters adjusted for the project, it was possible to fabricate the device with 25% internal infill. Figure 26 shows

an image of the printing process of the main components of the device generated by the printing software.

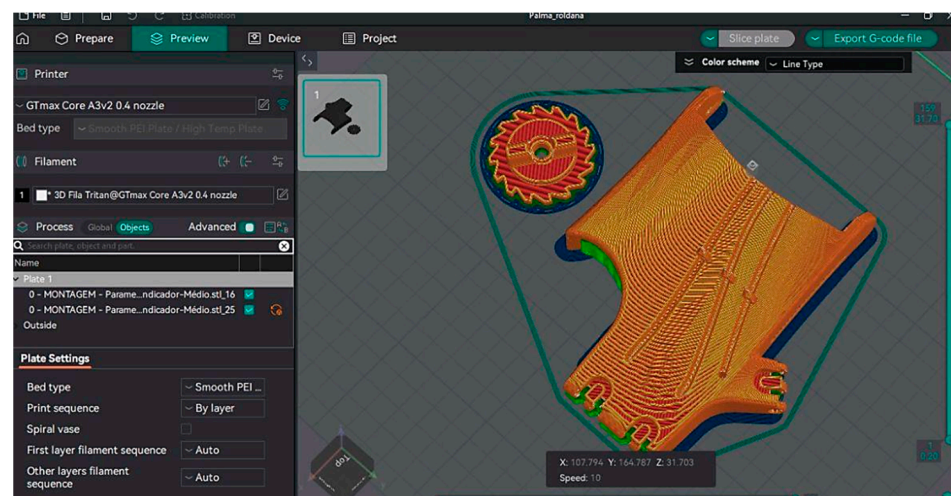


Figure 26. Software-generated image for initiating the three-dimensional printing process.

The parts fabricated through three-dimensional printing exhibit dimensional variations that directly depend on the selected printing plane and the geometric characteristics of the model. The choice of the YXZ printing plane is recognized for providing greater dimensional accuracy and better surface finish, ensuring greater uniformity in the printed components [25,26]. This approach reduces inconsistencies during the material deposition process, which is essential for applications requiring high quality and reliability [39].

By using the YXZ plane, it was possible to produce prosthetic parts with reliable dimensions and well-controlled surface roughness values, remaining below 23 μm . This configuration is especially advantageous for ensuring a precise fit between the parts and uniform adhesion of the layers. In the case of prostheses, this manufacturing technique ensures that the device meets functional and ergonomic requirements, allowing the user to use it comfortably and efficiently, as reported in the specialized literature.

3.8. Device Building

The prototype required approximately 6 h to print the polymer components, which weighed 134 g (Figure 27), while the assembly and finishing process took about 3 h.

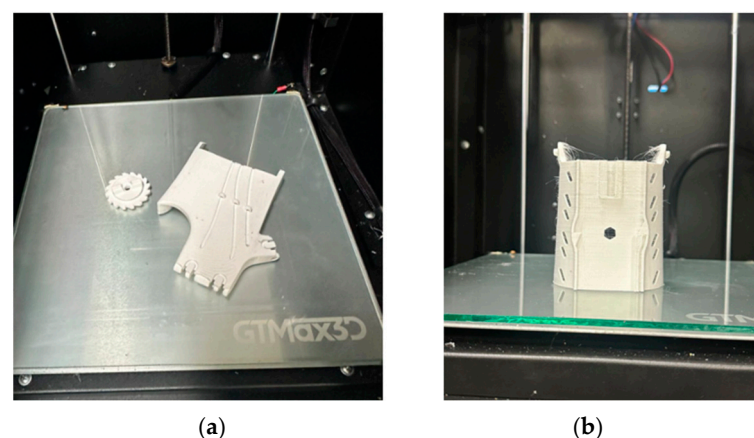


Figure 27. Images of (a) printing the palm and directional coupling and (b) printing the wrist.

The fully assembled device (Figure 28) reached a final weight of 240 g, an essential factor for the user adaptation process. The combination of mechanical strength and low

weight is crucial in the development of prosthetic devices, as it significantly contributes to efficiency and comfort during rehabilitation. In this context, the choice of a polymeric material like Tritan in the 3D printing process proved to be ideal.

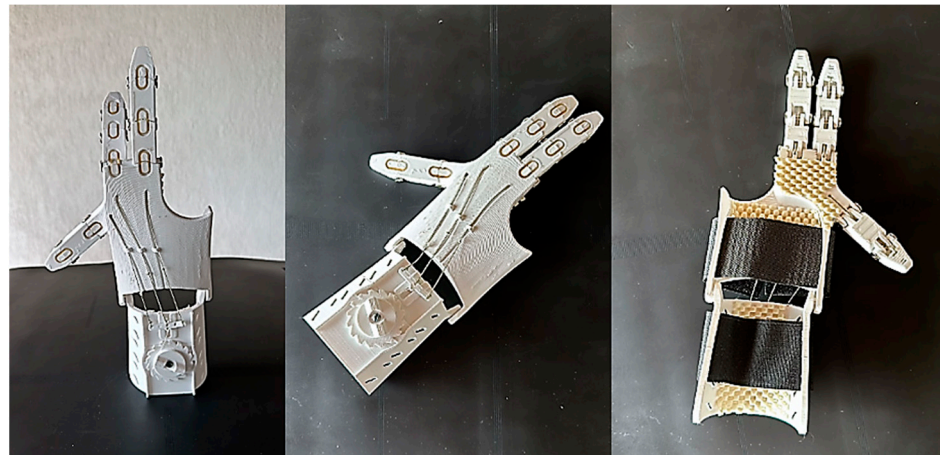


Figure 28. Images of the constructed device model.

Tritan, in addition to being a lightweight material, allows for manufacturing with 25% infill, resulting in an effective density of approximately 1 g/cm^3 . Its mechanical properties include a yield strength of 13.79 MPa, a maximum stress of 24.29 MPa, and an elasticity modulus of 332.54 MPa, characteristics that ensure a balance between lightness and robustness, making it an excellent choice for functional and comfortable prostheses.

The stiffness of the material can be evaluated through the value of the modulus of elasticity, while the initial plastic deformation value can be assessed by the yield strength or limit. These values demonstrate that it is feasible to use Fused Deposition Modeling (FDM) technology in the fabrication of hand prostheses, providing the orthopedic device with good mechanical strength and lightness.

After assembly, the prosthetic device was designed with the potential to be installed on the amputated limb using Velcro stabilizers with cushioned medical foam, protective sleeves, and tension adjustments made through a directional ratchet. These elements were planned to ensure comfort, stability, and functionality during daily use. The ratchet allows for the adjustment of the prosthesis movement through the tensioning of cables, while high-strength lines on the upper surface enable gripping movements at wrist flexion angles of 20° to 30° . The calibration of these movements can be performed with a goniometer, which measures the angles of the phalanges and wrist, ensuring precision and efficiency.

The mechanical functioning of the prosthesis is enabled by the use of high-density orthodontic elastics positioned on top of the interphalangeal joints, allowing for passive extension of the fingers. Finger flexion is achieved through high-strength cords fixed along the palmar surface and phalanges, activated by wrist flexion at angles of 20° to 30° . This configuration enables coordinated movement, where the wrist acts to flex the fingers toward the palm, providing functionality and precision in the movements.

This work focused on the development of the engineering aspects of the prosthesis, including material characterization and selection, design project, mechanical feasibility study through finite element analysis, and the development of parameterization for manufacturing. The prosthesis was designed to operate mechanically through muscular propulsion, incorporating elements such as directional coupling to adjust the biomechanical alignment of the prosthesis, high-density cords, screws, and other components that contribute to its functionality. Additionally, the material used in the prosthesis was chosen for its biocompatibility, ensuring that it does not cause allergic reactions on the user's skin. The prosthesis

was designed to provide maximum functionality for future patients, taking into account adaptation to each individual's anatomy and needs.

Although the prosthesis development was completed, it is important to emphasize that this study did not include any clinical validation phase. To ensure that the device can be properly installed during clinical validation, a Velcro fastening system was developed and will be incorporated in the subsequent phase of the process. Clinical validation will be conducted by the multidisciplinary team at the Bioengineering Laboratory (LABBIO) at UFMG, where this phase will take place. The present study focused solely on the development and engineering analysis of the prosthesis, as described above.

This device allows for adaptable gripping movements to hold objects of different shapes and sizes. It is capable of handling items such as a 500 mL PET bottle of water, a Fuji apple with an approximate diameter of 80 mm, a cube with 10 mm edges, cylinders from various screwdrivers, and a whiteboard marker with dimensions of 125 mm in length and 15 mm in width. Additionally, the device supports lifting other objects with varying weights, as long as they do not exceed the maximum limit of 2 kg, ensuring functionality and versatility in daily use.

The estimated cost of the hand prosthesis is approximately BRL 300.00 (USD 50.15), representing a low-cost option compared to conventional passive and active mechanical prostheses, which range from BRL 1500.00 (USD 254.74) to BRL 5790.00 (USD 984.26) [21]. In addition, upper limb prostheses with advanced operating systems on the global market can cost between USD 30,000 and USD 65,000 per unit [40,41].

The cost estimate for the prosthesis developed in this study was based on a detailed analysis of several factors, including the materials used, the manufacturing process, and associated operational costs. The polymer filament used for the 3D printing of the prosthesis was the main cost component, priced at BRL 230.00 (USD 38.50) per kilogram. Considering the total weight of the prosthesis is 240 g, the material cost per unit was calculated at approximately BRL 51.23 (USD 8.60). This value represents a significant portion of the total production cost, as the selected material offers good mechanical strength and lightness, which are essential for the prosthesis's proper performance.

In addition to the material costs, the 3D printing process itself is another relevant factor in the cost estimate. The prosthesis was manufactured using Fused Deposition Modeling (FDM), a process that, while efficient and cost-effective, incurs expenses for printer maintenance and energy consumption. The printing time cost was calculated based on energy consumption rates and the number of hours required to produce the prosthesis, resulting in an approximate value of BRL 15.00 (USD 2.55) per unit.

Other operational costs include auxiliary tools, such as modeling software, which allowed the prosthesis design to be customized for different amputation levels, and the use of surface finishes to ensure a proper and comfortable finish for the user. These additional costs were estimated at around BRL 15.00 (USD 2.55) per unit.

Compared to traditional prosthesis alternatives, the additive manufacturing approach offers a significantly more affordable solution, with the added advantage of being customizable for different amputation types and functional levels. This reduced cost allows the prosthesis to be a viable option for a greater number of users, providing an accessible solution without compromising on quality and mechanical efficiency.

The prosthetic device combines lightness, strength, and functionality, being accessible and efficient for rehabilitation. Manufactured in Tritan using 3D printing, it offers versatile performance at a reduced cost, standing out as an innovative and affordable alternative for hand prostheses.

4. Conclusions

The finite element analysis (FEA) conducted on the Type 3 partial hand prosthesis revealed significant results, confirming the feasibility of the current design. The maximum recorded stresses reached 33.456 MPa, with the highest stress concentrations observed at the finger joints, particularly in areas of contact with objects. These regions indicate the areas most susceptible to failure or deformation under continuous load. Although the prosthesis structure is adequate to withstand typical gripping loads, it is recommended to reinforce the joints to enhance the device's durability.

The displacement analysis of the prosthesis model showed a maximum deformation of 1.68 mm at the fingertips, indicating that the prosthesis undergoes deformation under the applied load conditions, especially in areas farthest from the fixation. This expected displacement emphasizes the importance of understanding the impacts of mechanical properties, such as maximum stress, yield strength, modulus of elasticity, and elongation, when selecting the polymer material for FDM printing. These properties are directly linked to the chemical composition and microstructural characteristics of the material, which are essential for ensuring good mechanical strength after three-dimensional printing.

Regarding the mechanical feasibility study, static load simulations performed with finite element software indicated that the prosthetic device is mechanically feasible for compressing solid objects under a 50 N force without experiencing plastic deformations or ruptures. The prosthesis model, developed with 25% infill and Tri-hexagonal internal fill geometry, showed good dimensional accuracy and surface finish, resulting in a device with low weight (240 g) and high mechanical strength.

The cost analysis revealed that the polymer filament is the most expensive component, priced at BRL 230.00 (USD 38.50) per kg. However, the prosthesis construction achieves an excellent cost-benefit ratio, totaling approximately BRL 300.00 (USD 50.15) per unit. Thus, it is possible to print and build a mechanical hand prosthesis with low production costs using additive manufacturing.

It is important to emphasize that this work focused on the development of the prosthetic device based on engineering principles, covering material selection, structural analysis, and mechanical system implementation. Clinical validation, involving human trials and real-world performance evaluations, will be a subsequent phase to be carried out in future stages of the project. Therefore, no clinical trials with the prosthesis have been conducted thus far, and clinical assessment should be carried out in future studies.

The prosthesis functions through a muscular propulsion system, where the movements of finger flexion and extension are controlled by forces generated by the remaining muscles of the amputee. High-density cords, fixed along the palmar surface and phalanges, activate finger flexion by being tensioned through wrist flexion. The directional coupling in the system also facilitates precise angulation of the prosthesis, enabling coordinated movement between the wrist and fingers.

In conclusion, the evidence obtained through the analyses and simulations confirms that the developed hand prosthesis not only meets functional requirements but also presents a viable solution in terms of cost and performance. The project has the potential to be improved in subsequent phases, with design enhancements and clinical validation, significantly benefiting users by combining quality, comfort, and durability.

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tration, R.C.d.S.R.; funding acquisition, C.B.S.V. All authors have read and agreed to the published version of the manuscript.

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